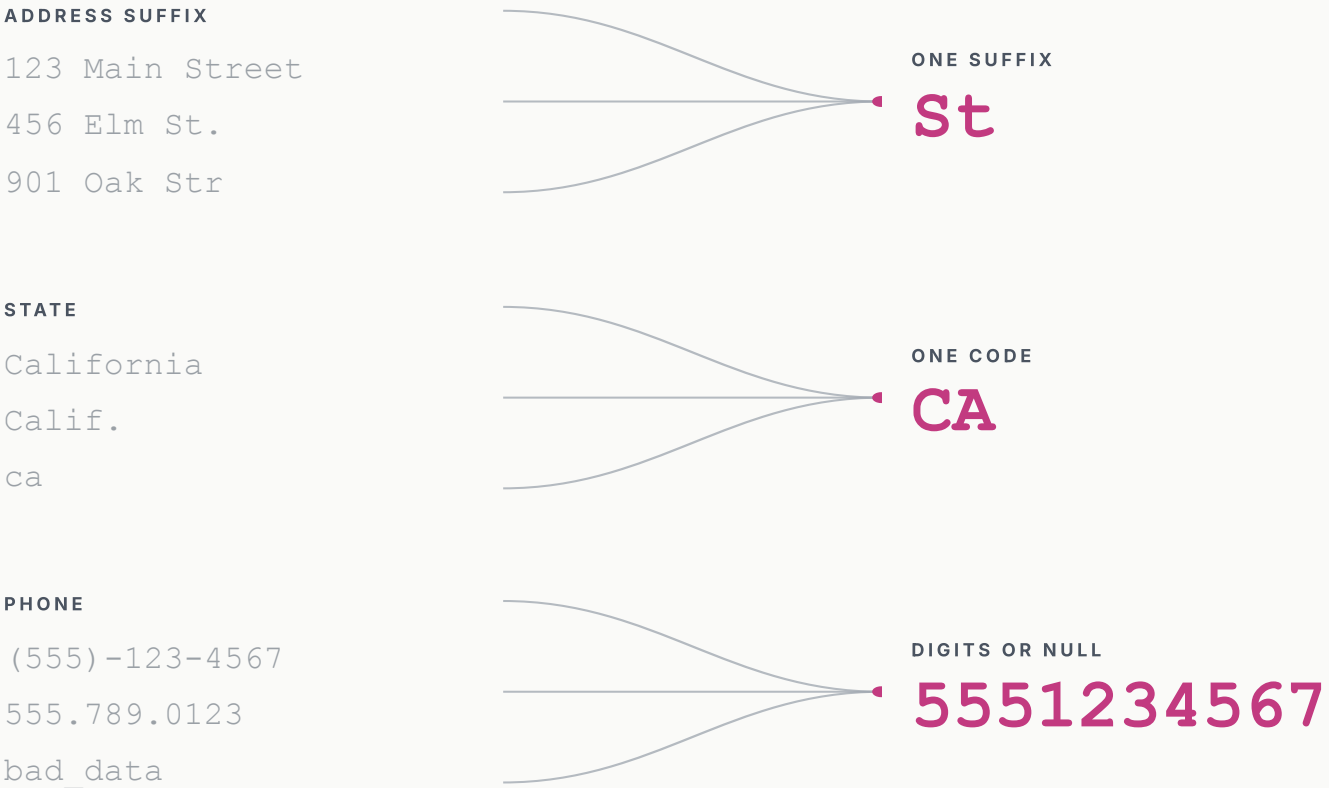


DATA ENGINEERING & ANALYTICS

TelcoPrime

Customer Data Trust Transformation.

A Microsoft Fabric case study in low-code transformation with Dataflow Gen2, reference data lookups, and address, state and phone standardisation for a telco operator rebuilding its customer foundation.

TELECOMMUNICATIONS · LAKEHOUSE · DATAFLOW GEN2
REFERENCE DATA · STANDARDISATION

MICROSOFT FABRIC

CASE STUDY 10 · CONTENTS

Five keys and a hash.

2,500 customer records. Eleven ways to write a street suffix. Thirteen ways to write four states. One phone number in nine that cannot be used. This is the brief.

1 p. 03

The Brief

Undeliverable mail, a churn model on bad features, and three options weighed.

2 p. 04

The Data

customer_bronze.csv, the two lookup files, and what is wrong in each column.

3 p. 05

The Contract

The Customer_Silver schema and the rules every row must follow.

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Lab & Acceptance

What you build in class, what you finish at home, and how each is checked.

5 p. 07

Discussion

Five questions on impact, tooling, transformation logic, scale and prevention.

#

Customer_Silver

The one table every team dials into.

SECTION 01 · THE BRIEF

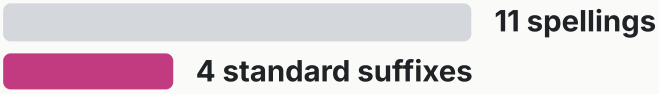
Clean the customer foundation.
Before the next campaign goes out.

TelcoPrime is building a Customer 360 view, but its customer records were never validated. Marketing mail comes back undelivered, and the churn model trains on location features it cannot trust. The Data Engineering Manager wants a low-code cleanse the team can run every day.

> A significant portion of direct mail campaign budget was wasted on undeliverable letters. The Customer 360 initiative has no value to marketing without clean, trusted address data. The requirement is standardised addresses, with urgency.
>
> DIRECTOR OF MARKETING, TELCOPRIME

THREE DEFECTS, MEASURED IN THE DATA

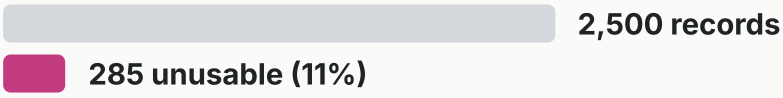
Street suffixes
Street, St., Str. and St all mean the same thing. Address matching fails.



States
California appears as CA, ca, Calif. and California. Location segments split four ways.



Phone numbers
Four formats, plus blanks and text such as bad_data and pending.



THREE OPTIONS, ONE RECOMMENDATION

Manual Excel cleanup REJECTED	FIXES: Quick for one file, no new tools.	LEAVES BROKEN: Repeated by hand every time. No two analysts produce the same mappings.
PySpark notebook REJECTED	FIXES: Scales to any volume; full control.	LEAVES BROKEN: The team does not write Python today. Nobody else can maintain it.
Dataflow Gen2 RECOMMENDED	FIXES: Low-code, visual steps the team can read and change. Runs daily.	LEAVES BROKEN: Heavier than code at very large volumes.

Only Dataflow Gen2 gives the team a cleanse it can read, own and schedule. The price is performance at very large volume, which a 2,500-row daily feed does not reach.

SECTION 02 · THE DATA

One CSV.
Two lookups that define the standard.

customer_bronze.csv sits in the Lakehouse Files area under CaseStudy10. The two lookup files beside it list every raw spelling found in the data and the standard value it maps to.

EXHIBIT A · SAMPLE FROM CUSTOMER_BRONZE.CSV

CUSTOMERID	1 ADDRESS	2 STATE	3 PHONE
1001	123 Main **Street**	**California**	**(555)-123-4567**
1002	456 Elm **St.** , Apt 4B	NY	**555.789.0123**
1003	789 Maple **Avenue**	**tx**	5551234567
1004	901 Oak **Str**	FL	**bad_data**

Highlighted values break the Customer_Silver contract.

EXHIBITS B AND C · REFERENCE LOOKUPS

state_lookup.csv
RAW_STATE → STANDARD_STATE

RAW_STATE	STANDARD_STATE	RAW_STATE	STANDARD_STATE
CA	CA	TX	TX
ca	CA	tx	TX
Calif.	CA	Texas	TX
California	CA	FL	FL
NY	NY	fl	FL
ny	NY	Florida	FL
New York	NY		

address_suffix_lookup.csv
RAW_SUFFIX → STANDARD_SUFFIX

RAW_SUFFIX	STANDARD_SUFFIX	RAW_SUFFIX	STANDARD_SUFFIX
Street	St	Ave	Ave
St.	St	Road	Rd
St	St	Rd.	Rd
Str.	St	Drive	Dr
Str	St	Dr.	Dr
Avenue	Ave		

DATA PROFILE

2,500
customer records, CustomerID 1001 to 3500.

518
addresses carry a unit after a comma, e.g. “456 Elm St., Apt 4B”.

168
military and PSC addresses (USS Cohen, PSC 5029) have no street suffix at all.

285
phones are blank, bad_data, not provided, pending or invalid.

SECTION 03 · THE CONTRACT

Six columns.
One table every team reads.

Customer_Silver is the contract marketing, analytics and engineering all consume. Column names and types are fixed. The tag on each column shows whether it is built in class or in the assignment.

EXHIBIT D · CUSTOMER_SILVER TARGET SCHEMA

CLASS	→ **CustomerID** int	Unique customer identifier.
CLASS	→ **Address_Street** string	Street address with a standard suffix (e.g. 123 Main St).
CLASS	→ **Address_Unit** string	Unit or apartment (e.g. Apt 4B), or NULL if absent.
CLASS	→ **Standard_State** string	Two-letter state code (e.g. CA).
ASSIGNMENT	→ **Phone_Clean** string	Ten digits only, or NULL if invalid.
ASSIGNMENT	→ **Data_Quality_Flag** int	1 for records meeting all quality checks. 0 for records with known issues.

RULES EVERY ROW MUST FOLLOW

01 Every state maps
Every raw state value appears in state_lookup.csv. No row may end with a NULL Standard_State.

02 Units follow a comma
Anything after the last comma in Address is the unit. Addresses without a comma have no unit.

03 Unknown suffixes survive
A street whose last word is not in address_suffix_lookup.csv keeps that word unchanged. Nothing is dropped.

04 No row is lost
Customer_Silver holds all 2,500 CustomerIDs from customer_bronze.csv.

SECTION 04 · LAB & ACCEPTANCE

One dataflow in class.
Four steps at home. One bonus.

Class builds the address and state cleanse. The assignment finishes the contract, lands the table and proves it with SQL. Each step is signed off by its own check.

IN CLASS

1 Dataflow Gen2

Reads customer_bronze.csv, merges both lookups, and builds Address_Street, Address_Unit and Standard_State.

DONE WHEN Every row has a Standard_State. 518 rows have an Address_Unit. Suffixes read St, Ave, Rd or Dr wherever the lookup knows them.

ASSIGNMENT

2 Phone_Clean

Strip every non-digit. Keep ten-digit results; everything else becomes NULL.

DONE WHEN 2,215 ten-digit phones. 285 NULLs. No other lengths.

3 Data_Quality_Flag

1 when a record meets all quality checks, 0 when it has a known issue.

DONE WHEN Every row is 1 or 0. No NULLs.

4 Customer_Silver table

Final column names, order and types, written to the Lakehouse and scheduled daily.

DONE WHEN Six columns exactly as the contract defines. 2,500 rows.

5 SQL validation script

Three checks on the SQL endpoint: suffixes standardised, phones numeric or NULL, flag populated.

DONE WHEN All three checks pass.

BONUS

6 Unmatched report

List the addresses whose last word is not in the suffix lookup.

DONE WHEN 168 military and PSC addresses, and nothing else.

SECTION 05 · DISCUSSION

Five questions.
Each one a decision you must defend.

Answer each against the brief and the contract, not the product documentation.

Q1 **Business Impact**

Marketing and the churn analytics team have different problems but the same root cause. Explain the business impact of dirty data to a non-technical executive without using technical vocabulary.

Q2 **Strategic Tooling**

Why is Dataflow Gen2 strategic for this initial transformation given the team's skillset and the maintainability requirement? What are the long-term benefits of establishing a low-code transformation pattern?

Q3 **Transformation Logic**

Walk through the logical steps required in Power Query to transform Address and Phone from Exhibit A to the Customer_Silver schema. Identify the key transformations and the order they must be applied.

Q4 **Scalability**

If volume grew to millions of records per day, how would this Dataflow Gen2 solution scale? What performance considerations must a data engineer account for when designing the pipeline on Microsoft Fabric?

Q5 **Prevention**

Cleansing is reactive. What validation in the source system would stop these defects arriving in the first place, and who should own the two lookup files?

From eleven spellings of Street to one table the business trusts.

WHAT YOU WILL DELIVER	A Dataflow Gen2 that standardises addresses and states in class. At home: Phone_Clean, Data_Quality_Flag, the Customer_Silver table on a daily schedule, and a three-check SQL validation script.
HOW YOU WILL BE EVALUATED	The Done-when checks on page 06, plus five discussion answers framed against the brief.
WHAT THIS PROVES	You can choose low-code deliberately, not by default. You can apply reference data joins to resolve a finite set of canonical mappings. You can engineer a quality flag that downstream consumers can filter against.

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